Accelerating Low-Frequency Convergence for Limited-Angle DBT

Two-Channel Fidelity in PDHG

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CT Meeting 2026

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Motivation

What is limited-angle reconstruction?

- Tomography recovers a cross-section from X-ray projections taken at many source angles around the object.
- **Sufficient sampling.** For a fan-beam scan, a source arc of 180° plus the fan angle already measures every line integral — a full 360° rotation only re-measures each ray from the opposite side.
- **Limited-angle reconstruction (LAR)** uses a source arc *below* that threshold. The measurements no longer determine the image: a wedge of spatial frequencies is never observed.
- Digital breast tomosynthesis (DBT) operates deep in this regime — a source arc of only $\sim 50^\circ$ — so its reconstruction is severely ill-posed.

Why limited-angle?

- **Dose** — fewer views means less ionizing radiation per scan.
- **Geometry** — in DBT the detector stays under the compressed breast while the X-ray tube sweeps a short arc; a full gantry rotation is mechanically impossible.
- **Acquisition time** — a short sweep is fast, limiting motion blur and patient discomfort.
- **Use cases** — DBT for early breast-cancer detection; also lung-nodule and kidney-stone imaging.

[DBT geometry schematic
50° source arc,
detector, breast cross-section]

Two reconstruction settings: 2D and 3D

The two-channel method is evaluated in both geometries:

- **2D fan-beam** — a single slice; cheap enough to sweep image resolution and noise level.
- **3D cone-beam** — the actual DBT geometry; full-volume reconstruction on anatomical phantoms.

Both are severely under-determined: the number of unknowns far exceeds the number of measurements, increasingly so at finer resolution.

Slice resolution	Unknowns (pixels)	Measurements (rays)
512^2	262,144	25,600
256^2	65,536	25,600
128^2	16,384	25,600

Measurements $=$ 25 views $\times$ 1024 detector bins.

The reconstruction setup

Reconstruction with PDHG (Chambolle–Pock)

The image is found by minimizing directional-TV and ℓ_1 sparsity, subject to the projections matching the data within a tolerance, with $f \geq 0$.

- **Why not gradient descent?** The objective is non-differentiable (ℓ_1, TV) and hard-constrained — gradient descent needs a smooth, unconstrained objective.
- **PDHG** handles this class with cheap projections and proximal maps, no derivatives. It keeps a primal variable (the image) and a dual variable (the data residual), and alternates:
  - a **dual step** — pull the data residual toward the tolerance ball (step σ);
  - a **primal step** — pull the image toward sparsity and $f \geq 0$ (step τ).
- τ and σ are coupled through the operator norm $\|K\|$; we set them by an empirical balance heuristic (appendix).

Single-channel data fidelity

Prior work[1] solves a constrained DTV / ℓ_1 problem with a square-root Hanning filter on the sinogram residual:

$$\min_{f \geq 0} \; \alpha_x \|\partial_x f\|_1 + \alpha_y \|\partial_y f\|_1 + \alpha_z \|\partial_z f\|_1 + \beta \|f\|_1$$

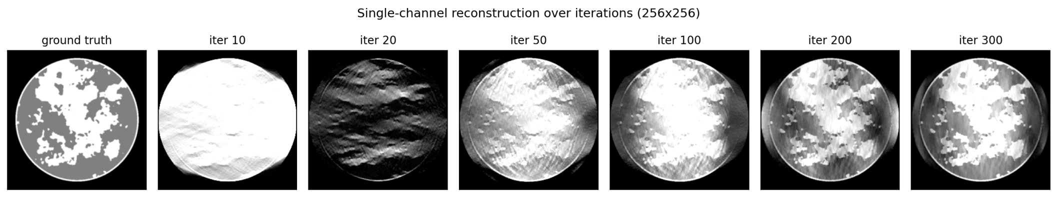
$$\text{s.t.} \quad \|R_{[c]}(g - Xf)\|_2 \leq \varepsilon \sqrt{|g|}.$$

- $R_{[c]} = \mathrm{Hann}^{1/2}(c)$ attenuates low spatial frequencies in the residual — noise suppression by design.
- Solved with PDHG and He–Yuan over-relaxation ($\rho \approx 1.75$).

[1] E. Y. Sidky et al., *J. Med. Imaging* 12(S1):S13013, 2025.

The problem

What happens during a single-channel run



- Fine structure (calcifications, glandular speckling) settles fast.
- Large soft-tissue regions stay blurred, with a slowly-varying intensity error, through iter $\sim$100–300.
- This is the low-frequency convergence bottleneck.

Why does this happen?

In the detector-frequency domain, the data-fidelity weight is

$$W(g - Xf), \qquad W = R_{[c]} = \mathrm{Hann}^{1/2}(c).$$

- PDHG corrects mode k at a rate $\propto s_k^2$ (squared singular values of WX).
- In LAR DBT, s_k is already tiny for low k (ill-conditioning).
- $\mathrm{Hann}^{1/2}$ shrinks low-k further *by design*.
- Stability bound is dictated by the *largest* s_k (high frequencies).

Consequence
The optimizer is forced to use step sizes that under-drive the very modes that are hardest to recover.

The idea: two-channel fidelity

Split the sinogram into two bands

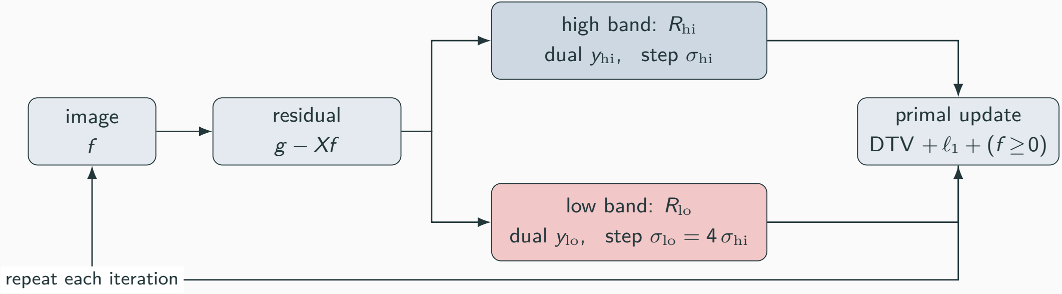
$$\min_{f\geq 0} \ \alpha_\bullet \|\partial_\bullet f\|_1 + \beta\|f\|_1$$

$$\text{s.t.} \quad \begin{cases} \|R_{\mathrm{hi}}(g - Xf)\|_2 \leq \varepsilon_{\mathrm{hi}}\sqrt{|g|}, \\ \|R_{\mathrm{lo}}(g - Xf)\|_2 \leq \varepsilon_{\mathrm{lo}}\sqrt{|g|}. \end{cases}$$

- $R_{\mathrm{hi}} = \mathrm{Hann}^{1/2}(c)$ – preserves the high-frequency channel from prior work.
- $R_{\mathrm{lo}} = \mathrm{Hann}^{1/2}(c_{\mathrm{lo}})$ with $c_{\mathrm{lo}} \gg c$ – a narrow-band low-pass channel that targets the slow modes.
- Independent dual variables $y_{\mathrm{hi}}, y_{\mathrm{lo}}$ with independent step sizes $\sigma_{\mathrm{hi}}, \sigma_{\mathrm{lo}}$.

Step sizes follow the same balance heuristic as the single-channel paper: $\tau\,\sigma_{\mathrm{hi}}\,\|K\|^2 \approx 1$, with $\sigma_{\mathrm{lo}} = 4\sigma_{\mathrm{hi}}$ to drive the slow modes harder.

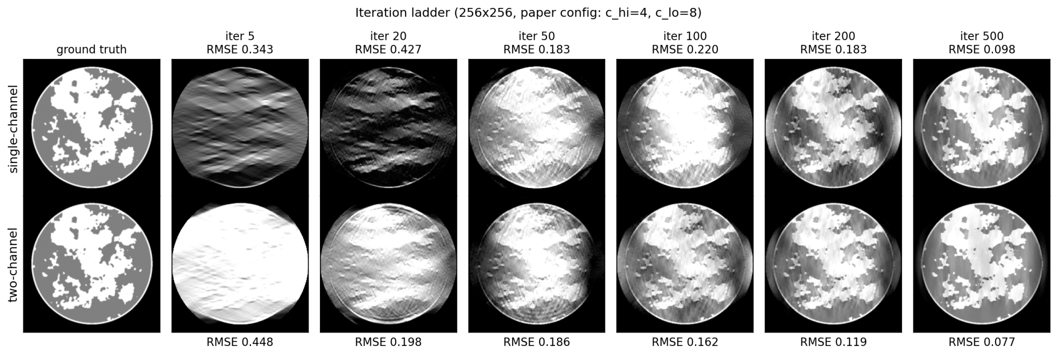
The two-channel iteration



- The data residual is split into a high- and a low-frequency band.
- Each band carries its own dual variable and its own step size.
- The low band is driven $4\times$ harder $(\sigma_{\mathrm{lo}} = 4\,\sigma_{\mathrm{hi}})$ — the slow modes receive the larger correction.
- The primal update and the image itself are unchanged from single-channel.

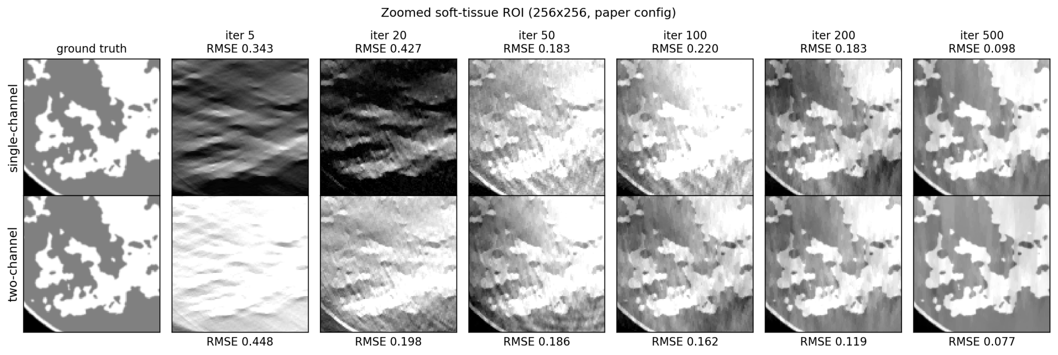
Results

Iteration ladder — single vs. two-channel



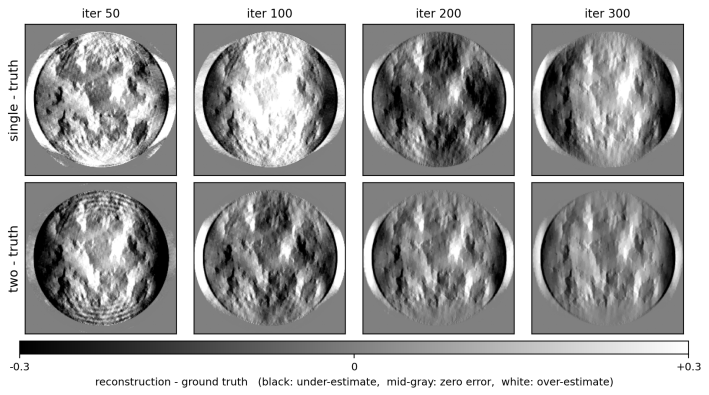
- 256×256, 25 views over 50°.
- Two-channel resolves recognizable soft-tissue structure several iterations earlier.

Soft-tissue ROI: low-frequency error



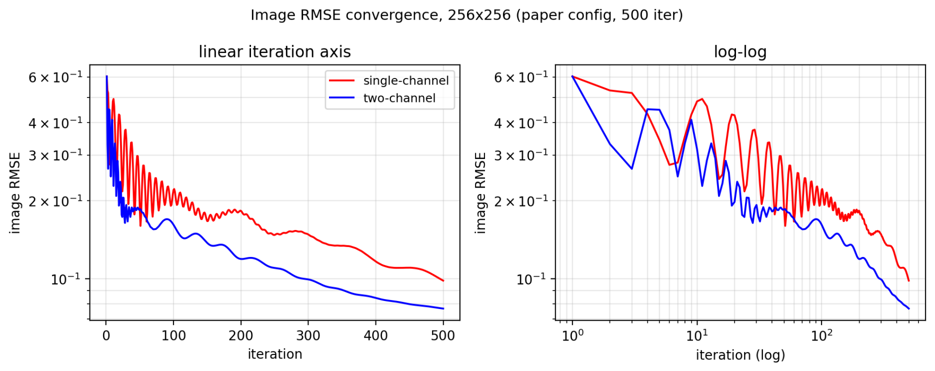
Iteration ladder cropped to a soft-tissue ROI. Single-channel carries a slowly-varying intensity error through iterations 50–200; two-channel reaches a uniform soft-tissue level several iterations earlier.

Where is the error?



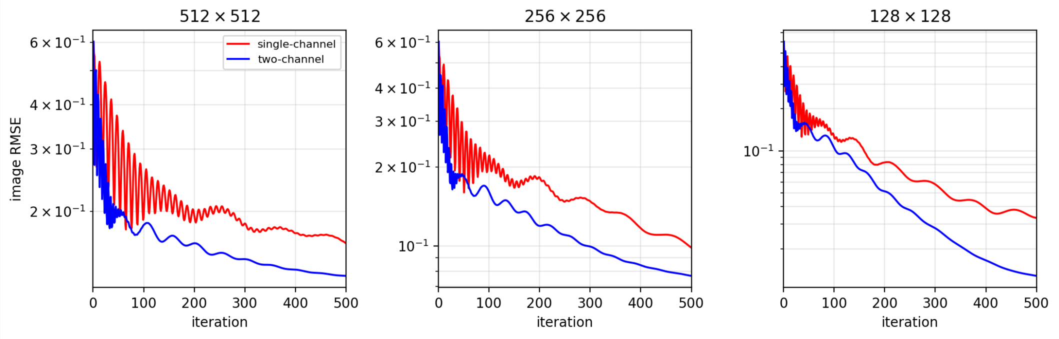
Difference image, reconstruction − ground truth (scale bar below). Single-channel at iter 100–200 carries a broad bright low-frequency bias; two-channel stays near uniform gray.

Convergence behaviour



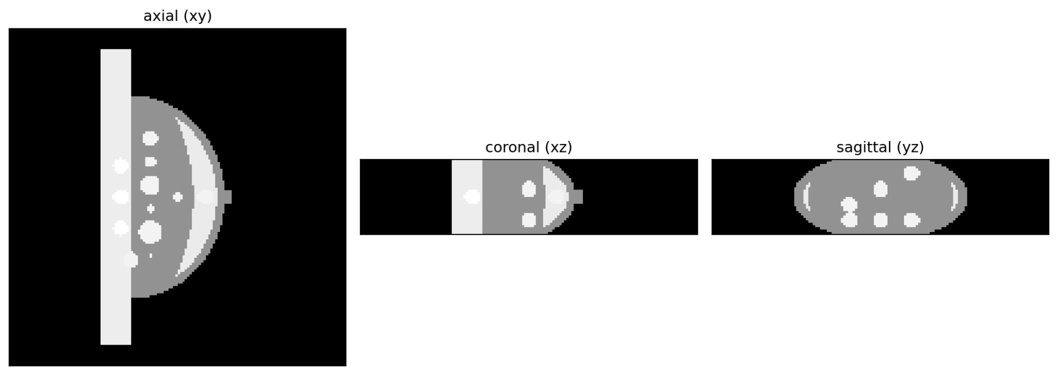
- Single-channel oscillates strongly through iter ∼200 — the cross-band interference predicted by the spectral analysis.
- Two-channel converges smoothly from the start.

Persistence across image sizes



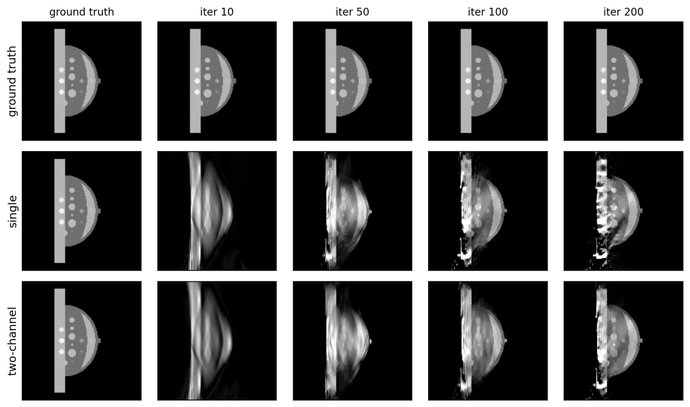
- Two-channel sits below single-channel across all three grids.
- Iter-500 RMSE reduction: 19% at 512^2, 22% at 256^2, 62% at 128^2.
- Consistent across the resolution sweep.

3D: Analytic breast phantom



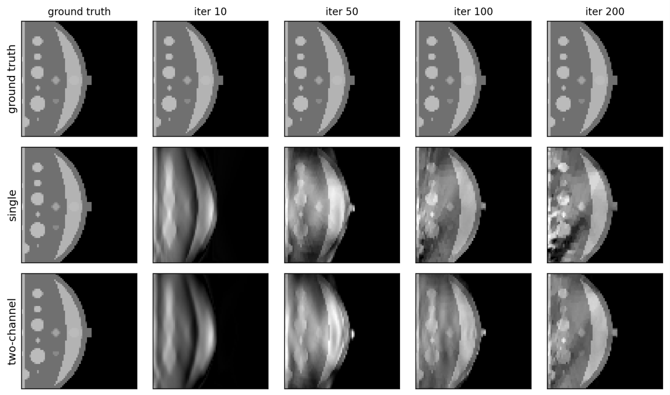
Analytic breast phantom: fatty soft-tissue background with discrete fatty-mass inserts and a chest-wall region. $144 \times 144 \times 32$ grid at 1.5 mm isotropic voxels.

Analytic breast — mid-axial iteration ladder



Cone-beam LAR (25 views / 50°), mid-axial slice. Two-channel image RMSE is 8% lower at iter 20, rising to a 35% reduction at iter 200; it resolves the discrete masses earlier.

Analytic breast — soft-tissue ROI

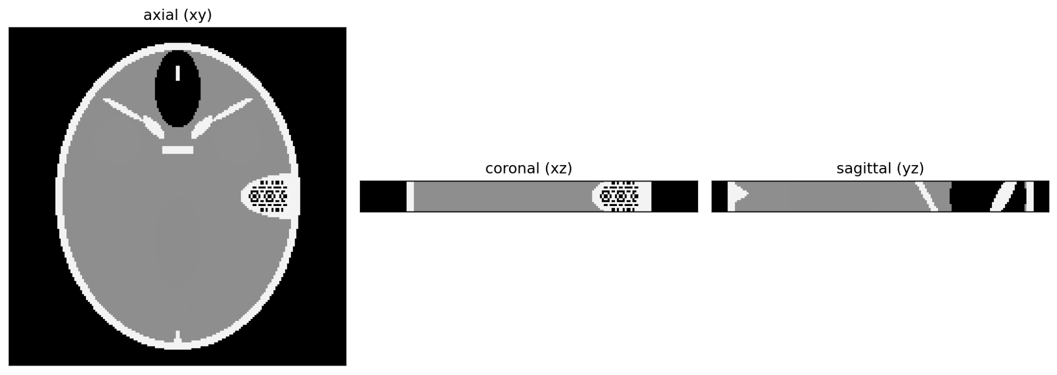


Lateral soft-tissue ROI with several fatty-mass inserts. Image RMSE at iter 100: single 0.077 vs. two-channel 0.051 — a 34% reduction. Two-channel shows discrete mass contours while single-channel retains a diffuse intensity bias.

Analytic breast — convergence

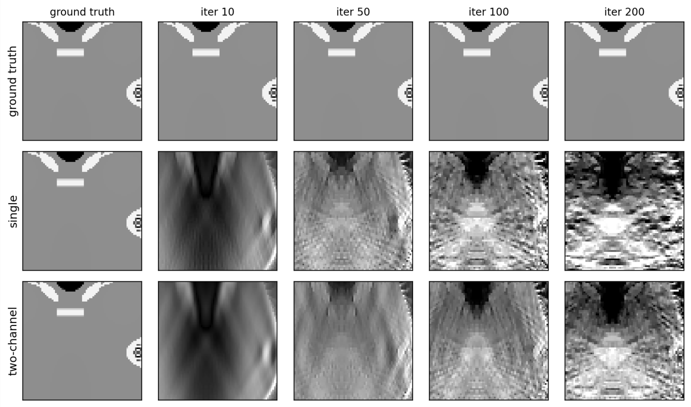
Image RMSE vs. iteration. Two-channel leads throughout: a 34% reduction at iter 100, peaking near 35% at iter 200 and holding 16% at iter 500.

3D: Analytic head phantom



Analytic head phantom: skull shell with sinuses, brain with ventricle and FORBILD low-contrast spots, subdural hematoma. 26 cm FOV, thin axial slab.

Analytic head — central brain ROI



Central brain region (low-contrast FORBILD spots, ventricle). Both methods drift upward under the under-determined geometry; the convergence plot makes the comparison quantitative.

Analytic head — convergence

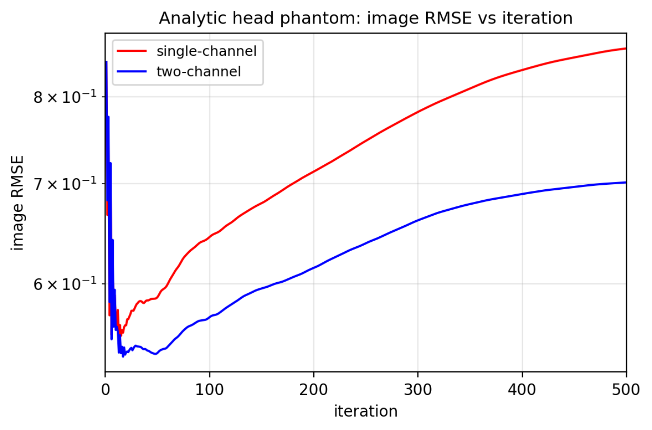
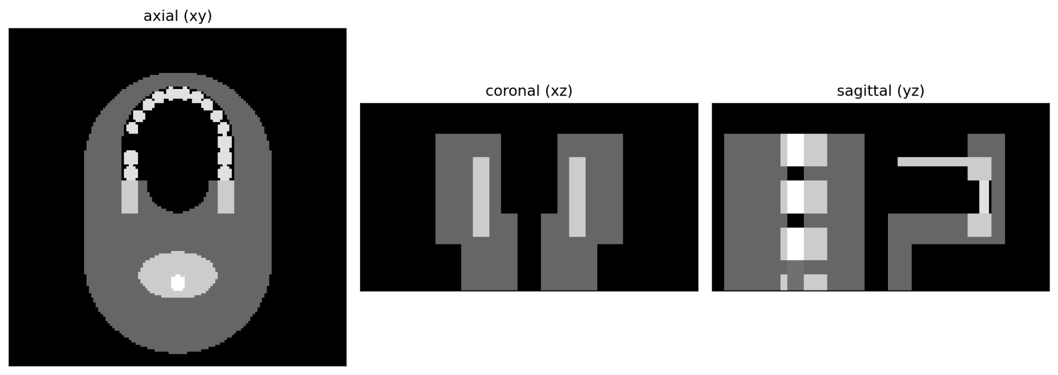


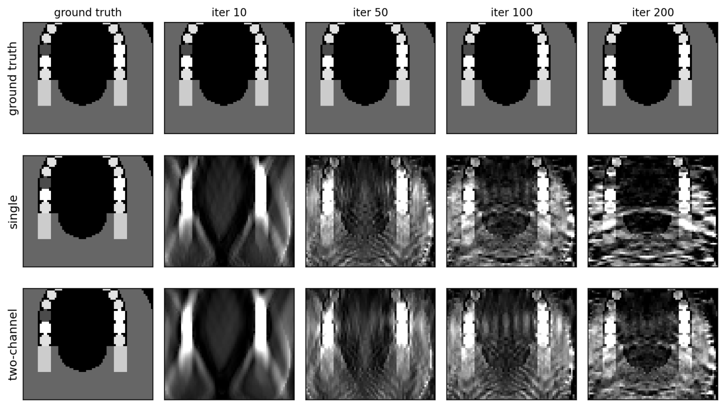
Image RMSE vs. iteration. Two-channel stays below single-channel from early iterations onward; the gap widens from 12% at iter 100 to 19% at iter 500.

3D: Analytic jaw phantom



Analytic jaw phantom: mandible/maxilla with 28 teeth, three gold crowns (high-μ scatterers), low-contrast tongue tumour. 12 cm tall volume.

Analytic jaw — mouth/teeth ROI



Mouth/teeth region. The gold crowns ($\mu = 19.3$) dominate the projection energy, leaving less low-frequency structure for the band split to act on; the convergence plot quantifies the gap.

Analytic jaw — convergence

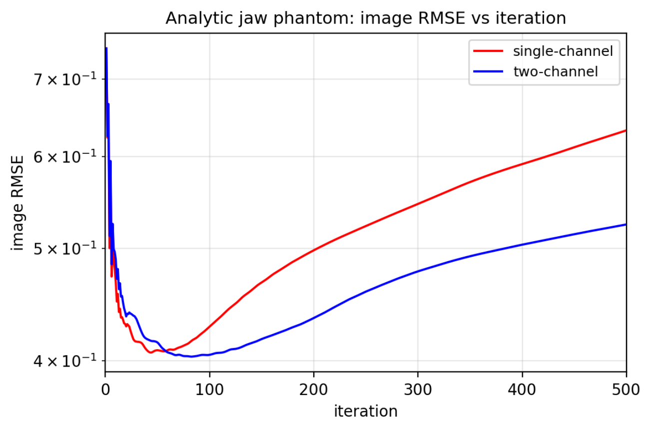


Image RMSE vs. iteration. Two-channel converges to a lower RMSE; the gap grows from 5% at iter 100 to 17% at iter 500.

Conclusion

Summary

- LAR DBT reconstruction has a *mode-dependent* convergence bottleneck: low spatial frequencies are starved of corrective drive because the PDHG step size is bound by high-frequency stability.
- Splitting the data fidelity into a low-pass and a high-pass channel, each with its own dual variable and step size, accelerates low-frequency settling without breaking stability.
- Consistent improvement at clinical iteration counts: 2D fan-beam across image sizes, and 3D cone-beam on the analytic breast, head, and jaw phantoms.

Future work

- **Multi-band fidelity.** Generalize from two channels to a geometric sequence of band-pass filters with matched per-shell step sizes. Manuscript in preparation.
- **Theorem-compliant step sizes.** Derive step sizes from a weighted power iteration so the empirical balance heuristic connects to the product-space PDHG convergence guarantee.
- **Realistic voxelized anatomy.** On the VICTRE breast phantom the two-channel gain at a clinical 50° arc is front-loaded — present early, then overtaken; sustaining it is ongoing work.

Questions?

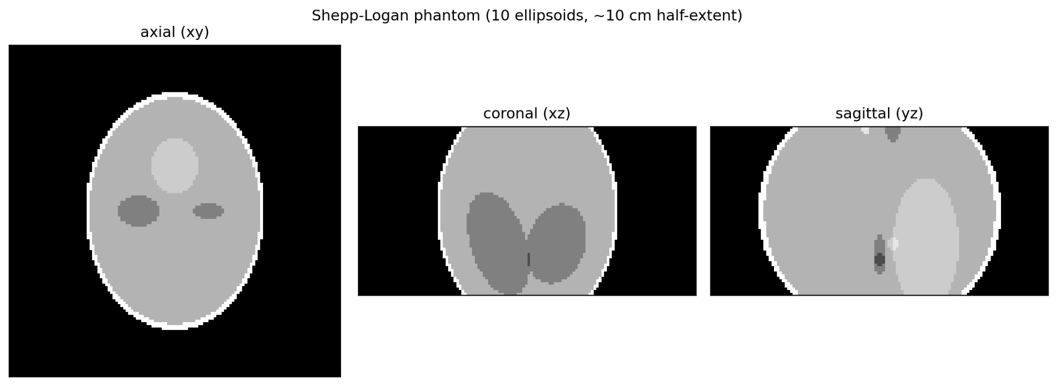
Backup: 3D phantoms at a clinical 50° arc

Two-channel vs. single-channel image-RMSE reduction (%) on the two realistic 3D phantoms, 25 views over 50°. Positive favours two-channel.

Phantom	Step-size config	iter 100	iter 200	iter 300	iter 400
Shepp-Logan	paper config	$+0.3$	-5.2	-6.5	-1.0
	tuned (best)	-1.5	-3.2	$+0.1$	-1.6
VICTRE breast	paper config	-5.6	-11.2	-17.8	-22.4
	tuned (best)	$+3.3$	$+1.1$	-6.0	-12.5

- At a realistic 50° arc neither 3D phantom shows the persistent gain seen on the analytic phantoms.
- VICTRE: tuning the CP step sizes yields a front-loaded gain — faster early, overtaken after iter $\sim$200. Shepp-Logan is a tie: a well-conditioned negative control.
- "Tuned" reduces the 3D step-size handicap (σ-inflation $2.0 \to 1.5$); the gain grows as the arc narrows below 50°.

Backup: 3D Shepp-Logan phantom



Classical 3D Shepp-Logan: 10 ellipsoids, skull at $\mu = 2.0$, brain at $\mu = 1.0$, three small low-contrast inserts at $\mu = 1.0 \pm 0.02$.

Backup: Shepp-Logan at a clinical 50° arc

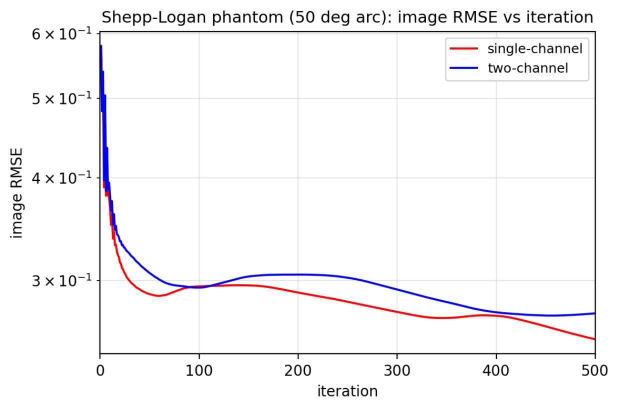
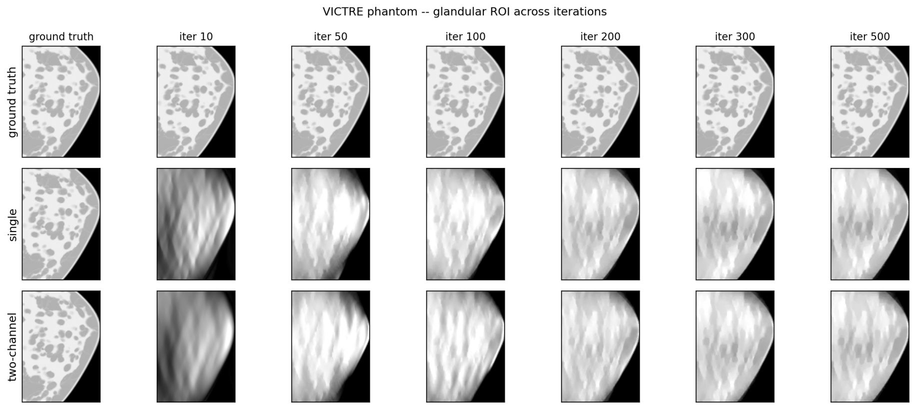


Image RMSE vs. iteration. Single- and two-channel are essentially superimposed (single is $\sim$7% lower at iter 500). With little intermediate spatial structure there is no low-frequency deficit for the spectral split to address — a well-conditioned negative control.

Backup: 3D VICTRE breast phantom

Voxelized VICTRE breast phantom (Reiser–Nishikawa anatomic model, $\sim 10^6$ voxels at 0.4 mm after $4\times$ downsampling). Anatomically distributed glandular structure rather than discrete analytic inserts.

Backup: VICTRE at a clinical 50° arc



Iteration ladder restricted to a glandular ROI; severe LAR streaking dominates both methods. Under the paper step sizes single-channel has the lower RMSE; tuning the CP step sizes produces an early two-channel lead that fades after iter $\sim$200 (see summary table).

Backup: full constrained objective

$$\min_{f \geq 0} \; \alpha_x \|\partial_x f\|_1 + \alpha_y \|\partial_y f\|_1 + \alpha_z \|\partial_z f\|_1 + \beta \|f\|_1$$

$$\text{s.t.} \quad \begin{cases} \|R_{\mathrm{hi}}(g - Xf)\|_2 \leq \varepsilon_{\mathrm{hi}} \sqrt{|g|}, \\ \|R_{\mathrm{lo}}(g - Xf)\|_2 \leq \varepsilon_{\mathrm{lo}} \sqrt{|g|}. \end{cases}$$

Parameter choices used for the figures in this talk:

	128^2	256^2, 512^2
α (DTV)	1.95	$1.9, 1.7$
β (ℓ_1)	10	$10, 5$
c_{hi}	4	4
c_{lo}	8	8
$\sigma_{\mathrm{lo}}/\sigma_{\mathrm{hi}}$	4	4
$\varepsilon_{\mathrm{lo}}/\varepsilon_{\mathrm{hi}}$	1.25	1.25

Backup: Algorithm 1 (PDHG with split fidelity)

Inputs: $g, X, R_{\mathrm{hi}}, R_{\mathrm{lo}}, \alpha_{x,y,z}, \beta, \varepsilon_{\mathrm{hi,lo}}, \sigma_{\mathrm{hi,lo}}, \tau, \rho$.
Initialize $f^0, \bar{f}^0 = f^0, y_{\mathrm{hi}}^0 = y_{\mathrm{lo}}^0 = 0$.
for $k = 0, 1, 2, \ldots$ **do**

$$
\begin{aligned}
y_{\mathrm{hi}}^{k+1} &\leftarrow \mathrm{Proj}_{B_{\mathrm{hi}}}\big(y_{\mathrm{hi}}^k + \sigma_{\mathrm{hi}} R_{\mathrm{hi}}(g - X\bar{f}^k)\big),\\
y_{\mathrm{lo}}^{k+1} &\leftarrow \mathrm{Proj}_{B_{\mathrm{lo}}}\big(y_{\mathrm{lo}}^k + \sigma_{\mathrm{lo}} R_{\mathrm{lo}}(g - X\bar{f}^k)\big),\\
f^{k+1} &\leftarrow \mathrm{prox}_{\tau\Phi}\big(f^k - \tau X^\top (R_{\mathrm{hi}}^\top y_{\mathrm{hi}}^{k+1} + R_{\mathrm{lo}}^\top y_{\mathrm{lo}}^{k+1})\big),\\
\bar{f}^{k+1} &\leftarrow f^{k+1} + \rho(f^{k+1} - f^k).
\end{aligned}
$$

Here Φ collects the DTV $+\ell_1+$ non-negativity terms.

Backup: Step-size heuristic (and future work)

These results use the paper's empirical balance heuristic:

$$\sigma_{\mathrm{hi}} = c_{\mathrm{bal}}/\|K\|, \quad \tau = 1/(c_{\mathrm{bal}}\|K\|), \quad \sigma_{\mathrm{lo}} = 4\sigma_{\mathrm{hi}},$$

with $\|K\|$ from an *unweighted* joint power iteration over $[R_{\mathrm{hi}}X; R_{\mathrm{lo}}X; \partial_x; \partial_y; \partial_z; \ell_1 I]$. The product-space CP theorem demands $\tau \cdot \|\Sigma^{1/2}K\|^2 < 1$, which the heuristic does not strictly satisfy when $\sigma_{\mathrm{lo}} > \sigma_{\mathrm{hi}}$.

- Empirically stable for 2D at 500 iters; in 3D we scale $\|K\|$ up by $\sqrt{\sigma_{\mathrm{lo}}/\sigma_{\mathrm{hi}}}$ for stability without changing the LF/HF ratio.
- He–Yuan over-relaxation $\rho \approx 1.75$ on all duals.
- Deriving theorem-compliant step sizes from a weighted power iteration is a clean direction for the journal extension.

Backup: full-resolution RMSE comparison

Image RMSE under the paper's empirical step-size setup (paper-config $c_{\mathrm{hi}} = 4$, $c_{\mathrm{lo}} = 8$, $\sigma_{\mathrm{lo}} = 4\sigma_{\mathrm{hi}}$):

	iter 200			iter 500		
Resolution	Single	Two	Δ	Single	Two	Δ
512×512	0.1953	0.1628	-16.7%	0.1625	0.1317	-19.0%
256×256	0.1829	0.1192	-34.8%	0.0984	0.0768	-22.0%
128×128	0.0827	0.0515	-37.8%	0.0332	0.0128	-61.5%

- Iter-500 RMSE reduction of 19/22/61% at $512^2/256^2/128^2$.
- Improvement is largest at 128^2; the iter-200 column shows the gap is broad across image sizes during the "clinical-iteration" window.
- Caveat: image RMSE is still a weak summary at this angular range – the visual ladder is the right primary reading.

References

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